



Store at: +2+8°C.

Presentation:

Cod. SU014LQ CONT: R1 1 x 30 mL. + R2 1 x 10 mL. + Cal.

Procedure

Quantitative determination of HDL Cholesterol.

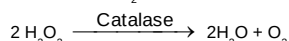
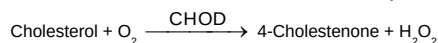
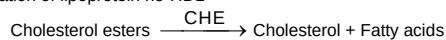
Only for in vitro use in clinical laboratory (IVD)

TEST SUMMARY

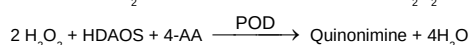
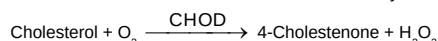
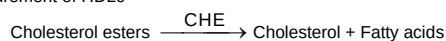
Directly determination of serum HDLc (high-density lipoprotein cholesterol) levels without the need for any pre-treatment or centrifugation of the sample.

The assay takes place in two steps.

- 1º Elimination of lipoprotein no-HDL



- 2º Measurement of HDLc



The intensity of the color formed is proportional to the HDLc concentration in the sample.

REAGENTS COMPOSITION

R.1	N,N-bis(2-hydroxyethyl)-2-aminoethanesulphonic acid pH 6.6	100 mM
	N-(2-hydroxy-3-sulfopropyl)-3,5-dimethoxyaniline (HDAOS)	0.7 mM
	Cholesterol Esterase	≥ 800 U/L
	Cholesterol oxidase	≥ 500 U/L
	Catalase	≥ 8300 KU/L
	Ascorbic oxidase	≥ 3000 U/L
R.2	N,N-bis(2-hydroxyethyl)-2-aminoethanesulphonic acid pH 7.0	1.1 mmol/L
	4 - Aminoantipyrine (4-AP)	100 mM
	Peroxidase	≥ 3500 U/L
HDLc/LDLc CAL	Lyophilized human serum.	

PRECAUTIONS

HDLc/ LDLc CAL

Components from human origin have been tested and found to be negative for the presence of HBsAg, HCV, and antibody to HIV (1/2). However handle cautiously as potentially infectious.

REAGENT PREPARATION AND STABILITY (Note 1)

R 1 and R 2: Are ready to use.

HDLc/ LDLc CAL: Dissolve the contents with 1 mL of distilled water. Cap vial and mix gently to dissolve contents.

Do not freeze the reagents.

HDLc/ LDLc CAL: Once reconstituted 30 hours at 20-25°C, 2 weeks at 2-8°C or 3 months at -20°C.

Do not use reagents over the expiration date.

Signs of reagent deterioration:

- Presence of particles and turbidity.

All the reagents of the kit are stable up to the end of the indicated month and year of expiry. Store tightly closed at 2-8°C. Do not use reagents over the expiration date.

SPECIMEN

Serum, heparinized plasma or EDTA plasma. If any samples show precipitates, centrifuge before using.

Stability of the sample: 6 days at 2-8°C and 1 year when stored at -70°C.

MATERIAL REQUIRED BUT NOT PROVIDED

- Spectrophotometer or colorimeter measuring at 570 nm.

General laboratory equipment.

TEST PROCEDURE

- Assay Conditions
 - Wavelength: 550-650 nm
 - Cuvette: 1 cm light path.
 - Temperature 37°C.
- Adjust the instrument to zero with distilled water.
- Pipette into a cuvette:

	Blank	Standard	Sample
R 1 (μL)	300	300	300
Calibrator (μL)	--	3	--
Sample (μL)	--	--	3

- Mix and incubate for 5 min at 37°C and read the absorbance (A₁) of the samples and calibrator.
- Add:

	Blank	Calibrator	Sample
R 2 (μL)	100	100	100

- Mix and incubate for 5 min. at 37°C and read the absorbance (A₂) of the samples and calibrator, against the Blank.

CALCULATIONS

$$\frac{(A_2 - A_1) \text{Sample} - (A_2 - A_1) \text{Blank}}{(A_2 - A_1) \text{Calibrator} - (A_2 - A_1) \text{Blank}} \times \text{Calibrator conc.} = \text{mg/dL of HDL-c in the sample}$$

Conversion factor: mg/dL x 0,0259= mmol/L.

QUALITY CONTROL

Control sera are recommended to monitor the performance of the procedure, H Normal and H Pathological (QC003, QC004).

If control values are found outside the defined range, check the instrument, reagents and calibrator for problems.

Serum controls are recommended for internal quality control. Each laboratory should establish its own Quality Control scheme and corrective actions.

REFERENCE VALUES

	Men	Women
Low risk	> 50 mg/dL	> 60 mg/dL
Normal risk	35 - 50 mg/dL	45 - 60 mg/dL
High risk	< 35 mg/dL	< 45 mg/dL

(These values are for orientation purpose).

It is suggested that each laboratory establish its own reference range.

CLINICAL SIGNIFICANCE

HDL particles serve to transport lipoproteins in the blood-stream.

HDL is known as "good cholesterol" because high levels are thought to lower the risk of heart disease and coronary artery disease.

A low HDL cholesterol levels, is considered a greater heart disease risk^{1,5,6}.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

REAGENT PERFORMANCE

- **Measuring Range:** From detection limit of 5.0 mg/dL. to linearity limit of 151 mg/dL., under the described assay conditions.

If results obtained were greater than linearity limit, dilute the sample ½ with NaCl 9 g/L. and multiply result by 2.

- **Precision:**

	Intra-assay (n=20)		Inter-assay (n=20)	
Mean (mg/dL)	28,0	76,1	28,0	76,1
SD	0,25	0,81	0,25	0,81
CV (%)	0,89	1,06	0,89	1,06

- **Accuracy:** Results obtained GPL (x) reagents did not show systematic differences when compared with other commercial reagents (y).

- The results obtained using 50 samples were the following: Correlation coefficient (r)²: 0.938

- Regression Equation : y=0.9825x - 1.41606

The results of the performance characteristics depend on the analyzer used.

INTERFERING SUBSTANCES

No interferences were observed to bilirubin. up to 30 mg/dL, hemoglobin up to 500 mg/dL, rheumatoid factors up to 1000 IU/mL or lipemia up to 1200 mg/dL.

Lipemic samples with a triglyceride concentration >1200 mg/dL should be diluted 1/10 with NaCl 9 g/L and multiply the result by 10.

NOTES

- The reagent 2 (R2) presents yellowish coloration due to the peroxidase, but it does not affect its functionality.

BIBLIOGRAPHY

- National Institutes of Health Consensus Development Conference Statement: Triglyceride, High Density Lipoprotein and Coronary Heart Disease. Washington D.C, Feb 26-28, 1992.
- Izawa S., Okada M., Matsui H., and Horita Y. J. Medicine and Pharmaceutical Sci., 1385 - 1388, 37 (1997).
- Shih WJ, Bachorik PS, Haga JA, Myers GL, Stein EA; Clinical Chemistry, 2000; 46:3:351 - 364
- Third Report of the National Cholesterol Education Programme (NCEP) Expert Panel on Detection, Evaluation and treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III). JAMA Publication, Vol 285, No. 19, P2486 - 2497; 2001.
- Jacobs, D. et al. In Laboratory and Test Handbook; Jacobs, D.S.; Kasten, B.L., De Mott, W.R., Wolfson, W.L., Eds; Lexi - Comp Inc: Hudson (Cleveland), 1990; P. 219.



CHEMELEX, S.A.
Pol. Ind. Can Castells. C / Industria 113, Nau J
08420 Canovelles -BARCELONA-
Tel- 34 93 849 17 35 Fax- 34 93 846 78 75

CHEMELEX, S.A.
Pol. Ind. Can Castells
C/ Industria 113, Nave J
08420 - CANOVELLES (Barcelona)
Tel- 34 93 849 17 35
Fax- 34 93 846 78 75

GPLBSDTT37-I
Rev. 9 - 16/05/19